

**SIGNSIGHT: EMPOWERING ACCESSIBILITY WITH
TAMIL SIGN LANGUAGE RECOGNITION USING
MACHINE LEARNING**

25-26J-404

Project Proposal Report

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Sri Lanka Institute of Information Technology

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August 2025

**EMOTION AND CONTEXT-AWARE TAMIL
SIGN LANGUAGE SUPPORT FOR CHILDREN**

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:



Date: 25/9/19

Signature of the co-supervisor:



25/09/19

Date:

Abstract

The majority of deaf children, including those who cannot speak, employ facial expressions and soft non-verbal sounds such as cries or murmurs for their main mode of communication. Teachers and parents would misinterpret these and have no idea whether the child is hurting, starving, sad, or angry. Current Tamil Sign Language (TSL) systems focus on hand movements with robot-like outputs that exclude emotions and non-verbal sounds.

This project offers an emotion- and sound-sensitive TSL module that is appropriate for children. The system identifies facial expressions via camera and microphone and classifies non-verbal sounds through computer vision and audio processing methods. A fusion engine combines these cues to identify the child's emotion and generates expressive TSL avatar animations with text overlays (e.g., "I am sad", "I am hungry"). This allows the caregivers to easily understand the needs of the child in a natural and unambiguous way.

The system also has a child-oriented interface with interactive drill and feedback in real-time, and learning is more participative and supportive. Through the use of gestures, emotions, and non-verbal sounds, this system aims to improve communication, emotional development, and confidence for Tamil hearing-impaired children of Sri Lanka.

Keywords: Tamil Sign Language, Emotion Detection, Non-Verbal Sound Analysis, Assistive Learning, AI for Children

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1 INTRODUCTION

Communication is an essential human need and the cornerstone of learning, socialization, and emotional growth. For deaf children, however, communication is a daily challenge. In Sri Lanka, the majority of the hearing-impaired children are communicating with their relatives, teachers, and peers in Tamil Sign Language (TSL). While TSL has an opening point for communication, there are important limitations—especially in expressing emotions and non-verbal information that are essential to natural understanding.

Most existing digital sign language systems can only identify hand motions. These devices may be beneficial for acquiring simple vocabularies but cannot decipher facial motions or silent sounds such as crying or murmurs, which best convey a child's actual sentiments. Thus, the child goes through mechanical, not complete, and emotionless communication. For children who do not speak, like those with cochlear implants, this chasm is even bigger, because their non-verbal sounds and emotions are easily misconstrued by parents and educators.

Current work in artificial intelligence (AI) shows the combination of vision and sound technologies facilitates improved recognition. Studies in Japanese Sign Language and American Sign Language show facial expressions function as grammar, separating statements from questions. Other research shows analyzing sound characteristics like loudness and pitch can determine that a cry is for pain, hunger, or frustration. Current technologies like MediaPipe, OpenCV, and Librosa already support real-time emotion and audio recognition in day-to-day applications.

Despite these developments, far less has been done for Tamil Sign Language, and little to nothing is directed towards children's particular needs. Existing systems lack adaptive, multimodal characteristics that recognize facial and audio signs as well as respond in a child-oriented way. To meet this, the proposed research develops an Emotion and Non-Verbal Sound-Aware TSL module for children. The system will

capture facial expressions and non-verbal sounds through a camera and microphone, classify the child's emotional state, and generate expressive TSL avatar animations along with text.

Synthesizing emotion, non-verbal sound, and context, the project aims to create an inclusive, adaptable, and child-friendly interface that not only improves communication but also enables emotional development, confidence, and social interaction for hearing-impaired children in Sri Lanka.

1.1 Background and Literature Survey

Something beyond the instruction of static hand signs needs to be accomplished in order to facilitate communication in a meaningful way among hearing-impaired children. Tone, emotion, and context are also part of communication, especially for children who cannot speak but communicate via crying, murmurs, or other non-verbal sounds. These aspects are not addressed by traditional sign language instructional materials, and therefore, interaction comes out looking mechanical and difficult for caregivers to interpret.

Over the last ten years, there has been research on computer vision and deep learning methods to improve the recognition of sign language. Bajwa and Gandhi [1] authored a systematic review of Indian Sign Language machine learning approaches, showing that convolutional and recurrent networks obtain consistent accuracy in gesture and facial expression recognition. However, they noted that emotional and contextual cues remain underexploited, limiting the naturalness of interaction for children. Similarly, Rosi et al. [2] employed deep learning for gesture recognition to support visually impaired users. While their work successfully recognized hand gestures, it did not address sign language as an entire communication system and neglected emotional and audio cues, highlighting the limitations of single-modality approaches.

In American Sign Language (ASL), some efforts have also combined gesture recognition and affective cues. Jamwal et al. [3] developed a system that translated ASL gestures to text and also incorporated basic emotion detection. Although their effort shows the potential of multimodal integration, it was limited to text outputs and might not be readily accessible to children. Significantly, the system did not include non-verbal sounds such as crying or murmurs that are important for children who cannot make spoken words.

Closer to the regional context, Fernando et al. [4] presented a system that incorporated Sinhala Sign Language with emotion recognition and audio recognition. Their results demonstrate the potential of using visual and audio channels for communication for

deaf children. Yet, this system was not tailored to Tamil Sign Language and not specialized in expressive non-verbal audio cues, leaving a significant gap for Tamil-speaking children in Sri Lanka.

Speech recognition for hearing-impaired children also warrants the inclusion of non-verbal audio. Chandrashekar et al. [5] recognized limited-vocabulary Kannada speech using spectrograms and CNN models. Their findings confirm that even unusual or impaired audio data can be processed effectively. This is proof that non-verbal children's sounds—such as pitch variations, murmurs, or cries—can also be modeled for improved emotional understanding.

Aside from speech and sign, multimodal models in child-oriented research are still proving the value of fusing inputs from more than a single source. Kurian and Tripathi [6] proposed m_AutNet, a personalized multimodal emotion recognition system for autistic children. By fusing facial expressions and vocal cues, their model significantly improved emotion detection accuracy, with performance much higher than unimodal models. Their findings confirm the importance of combining visual and audio information, offering direct justification to the motivation behind this work in combining facial emotions and non-verbal sounds for Tamil children.

Summary: From these studies, the state of the art demonstrates strong advancement in gesture recognition, multimodal emotion recognition, and speech processing. However, no existing system has been developed for Tamil Sign Language that combines facial emotion recognition and non-verbal sound analysis in a child-oriented model. This project bridges that gap by integrating these modalities to synthesize expressive TSL avatar animations with text so that children can communicate freely and caregivers can better comprehend them.

1.2 Research Gap

Despite many studies in sign language recognition and emotion detection being done, there is no system that combines facial emotion analysis and non-verbal sound detection into a single system for Tamil Sign Language (TSL). Most of the work that has been developed is for American Sign Language (ASL), Japanese Sign Language (JSL), or Chinese Sign Language (CSL). These systems are effective in gesture recognition and, in some cases, facial cues, but they do not address the unique linguistic and cultural aspects of Tamil.

More importantly, almost all current systems are not child friendly. Children require interactive, responsive, and emotionally engaging learning systems, but current systems are static in nature with a focus on rigid gestures and text output. They ignore that the majority of children, and especially non-verbal children, rely heavily on crying patterns, murmurs, and facial expressions to convey their needs. By failing to consider these cues, existing solutions leave caregivers questioning whether a child is sad, frustrated, or in pain.

Another major gap is the combination of multimodal cues. While some research has been done on the grammatical role of facial expressions or emotional content of audio signals, there is no TSL system that integrates face + sound + context to generate a clean and expressive output. Nor are there adaptive systems that respond to a child's mood or emotional state, with no support for engaging children's motivation and participation in real-world learning or communication settings.

As of now, there is a clear absence of an emotion-sensitive, sound-sensitive, and child-friendly TSL module that is capable of:

- Recognizing facial emotions in real time.
- Recognizing and classifying non-verbal cries and murmurs.
- Merging multimodal cues to generate expressive TSL avatar animations with text.
- Giving caregivers and educators real-time feedback that is clear.

This project bridges this gap by proposing an AI-powered TSL module that combines facial emotion detection and non-verbal sound into one, child-friendly module. It aims to create a natural, emotive, and compassionate communication experience never attempted in Tamil Sign Language research.

1.3 Research Problem

Sri Lankan children with hearing disabilities face extreme difficulties in expressing their feelings and needs through Tamil Sign Language (TSL). Current systems only cater to hand gestures and not facial expressions or non-verbal utterances like cries, murmurs, or whimpers. This leads to miscommunication because parents and instructors cannot understand if a child is sad, hungry, in pain, or frustrated. The questions below highlight the most crucial problems and how this project addresses them:

1. In what ways may both facial emotions and non-verbal sounds be recorded by systems?

Existing TSL systems are limited to identifying gestures only. The project integrates computer vision to identify emotions such as happy, sad, angry, and neutral, and audio processing to identify non-verbal sounds such as crying or murmurs.

2. How may these signals be combined into a coherent picture of the state of the child?

Most solutions treat vision and audio separately. This work proposes a Fusion and Mapping Engine that fuses facial expression outcomes with sound classification in order to determine the child's emotional state.

3. In what ways can the system express these states naturally to caregivers?

Outputs are currently mechanical and lack emotional content. The proposed system will generate TSL avatar animations with text descriptions (e.g., "I am sad", "I am angry"), which will render the child's state immediately clear and natural to parents and teachers.

Through these steps, this research explicitly addresses the gap in current systems by developing an Emotion & Non-Verbal Sound-Aware TSL module that provides children with an expressive, real-time, and child-friendly way of being understood.

2 OBJECTIVES

2.1 Main Objective

1. Develop a comprehensive learning platform for Tamil Sign Language (TSL):

- Create an interactive system for teaching TSL with structured lessons, assessments, and quizzes.
- Enable users to learn both static and dynamic signs, improving communication for the hearing-impaired community in Sri Lanka.

2. Enhance dynamic sign recognition using machine learning:

- Build a system capable of recognizing and interpreting dynamic TSL gestures using advanced computer vision techniques.
- Implement real-time recognition of gestures, improving accessibility and communication for TSL users.

3. Bridge the gap between spoken Tamil and TSL through audio/video conversion:

- Develop a tool that converts Tamil audio and video inputs into corresponding TSL signs.
- Provide an intuitive solution that helps hearing-impaired individuals understand spoken language via visual representation.

4. Develop emotion and context-aware TSL enhancement for children

- Design interactive exercises to help children, including those with cochlear implants, recognize and express emotions through TSL.
- Track progress and provide real-time feedback to improve expressive and communication skills in children.

5. Foster inclusivity and accessibility for the hearing-impaired community in Sri Lanka:

- Create a user-friendly platform that enhances communication, education, and interaction opportunities for the hearing-impaired
- Ensure that technology promotes social inclusion, making daily life and learning easier for TSL users.

2.2 Specific Objective

- **Collect and preprocess multimodal children's facial expression and non-verbal sounds data:**

The goal is to establish a dataset of happiness, sadness, anger, and neutral emotions expressed on faces and non-verbal sounds like crying, murmurs, or whimpers. The data will be cleaned, anonymized, and preprocessed to be of good quality for model training.

- **Apply computer vision for facial emotion detection:**

This involves the use of such software as MediaPipe and OpenCV to detect facial landmarks and extract features utilized for classifying children's emotions in real time. Such detected emotions will form an integral input when expressive TSL outputs are generated.

- **Design audio analysis for non-verbal sound classification**

This sub-goal uses audio processing techniques in Librosa and Praat to analyze pitch, loudness, and frequency structure of non-verbal sounds. The goal is to classify signals into emotional states of the child.

- **Combine multimodal outputs to make an inference of emotional state:**

A rule-based late-fusion decision system will be implemented to combine facial emotion and audio analysis outputs. This ensures that the final inferred emotional state is robust, reliable, and contextually appropriate.

3 METHODOLOGY

3.1 Technologies

Integrated development environment (IDE):

Development will be carried out using Visual Studio Code (VS Code). It is lightweight, user-friendly, and supports both Python (backend) and JavaScript (frontend) with useful extensions for debugging, testing, and GitHub integration.

Backend:

The backend will be implemented using Python (Flask) or Node.js, as both provide strong libraries for image/video processing, audio analysis, and decision-making logic. These frameworks allow smooth integration of multimodal inputs and real-time output generation.

Frameworks and Libraries:

- OpenCV – for preprocessing and handling image/video frames from the camera.
- MediaPipe – to detect and track facial landmarks for emotion classification.
- Librosa / Praat – to extract pitch, loudness, and frequency features from children’s non-verbal sounds (crying, murmurs, whimpers).
- Rule-based Fusion Engine – for combining emotion and sound cues to decide the child’s state.

Frontend:

The frontend will be built using React.js, offering a child-friendly interface with an animated avatar and text output. The design will use bright visuals, simple navigation, and interactive feedback to make the system accessible for children and caregivers.

APIs/Services:

- Text-to-Sign Animation API or custom 3D avatar modules to display expressive TSL outputs.
- Cloud Hosting (AWS / Google Cloud) to provide scalability, data storage, and real-time access.

Database:

- Firebase / Cloud Database – to store user data, feedback logs, and system outputs in a secure, encrypted format.
- Local Storage (JSON/SQLite) – for caching small sign animations and offline use.

Algorithms/Techniques:

- Facial Emotion Detection: Using MediaPipe and OpenCV to identify emotions such as happy, sad, angry, and neutral.
- Non-Verbal Sound Analysis: Using Librosa/Praat to analyze pitch (F0), loudness, and energy to classify crying/murmur sounds.
- Data Fusion: A rule-based decision-making engine will merge facial and audio outputs into a single interpretation of the child's emotional state.
- TSL Avatar Output: Selected emotions will be mapped to expressive TSL animations with matching text (e.g., *"I am sad"*, *"I am angry"*).

System Workflow:

1. Input Layer: A webcam and microphone capture the child's facial expressions and non-verbal sounds.
2. Processing Layer:
 - MediaPipe + OpenCV detect and classify facial emotions.
 - Librosa + Praat analyze pitch and loudness of crying/murmurs.
3. Fusion Layer: A mapping engine combines Emotion + Sound cues to determine the child's state.
4. Output Layer: The system displays an expressive TSL avatar animation with text labels.
5. Feedback Layer: Real-time feedback confirms whether the system understood correctly (e.g., "Correct / Try again").

3.2 System Architecture

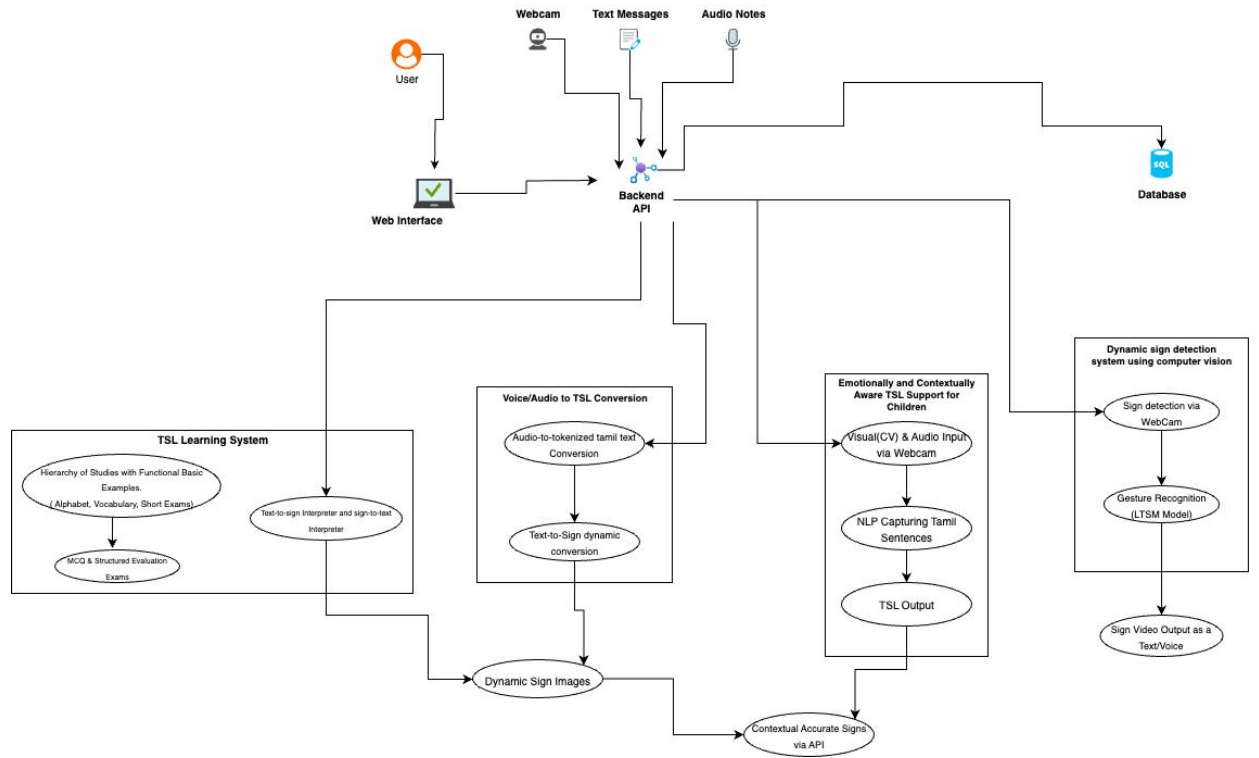


Figure 3.1 System Architecture

3.3 Component Overview Diagram.

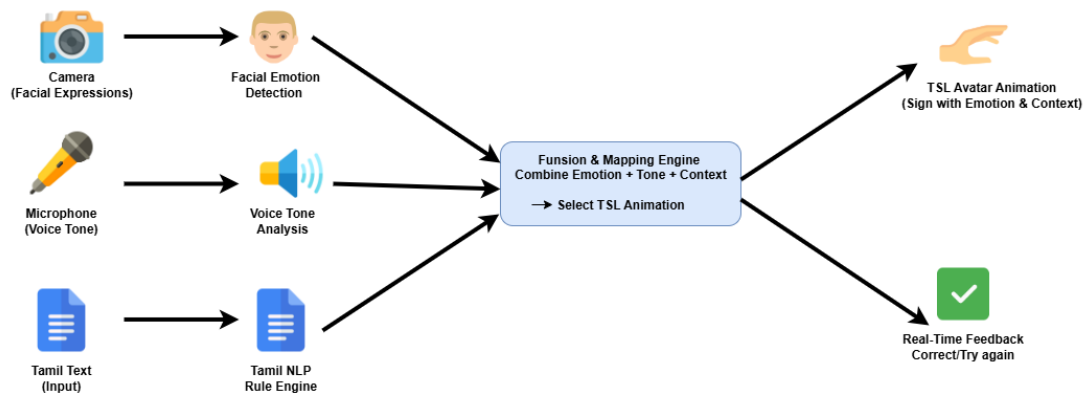


Figure 3.2 Component Overview Diagram

3.4 Work Breakdown Structure

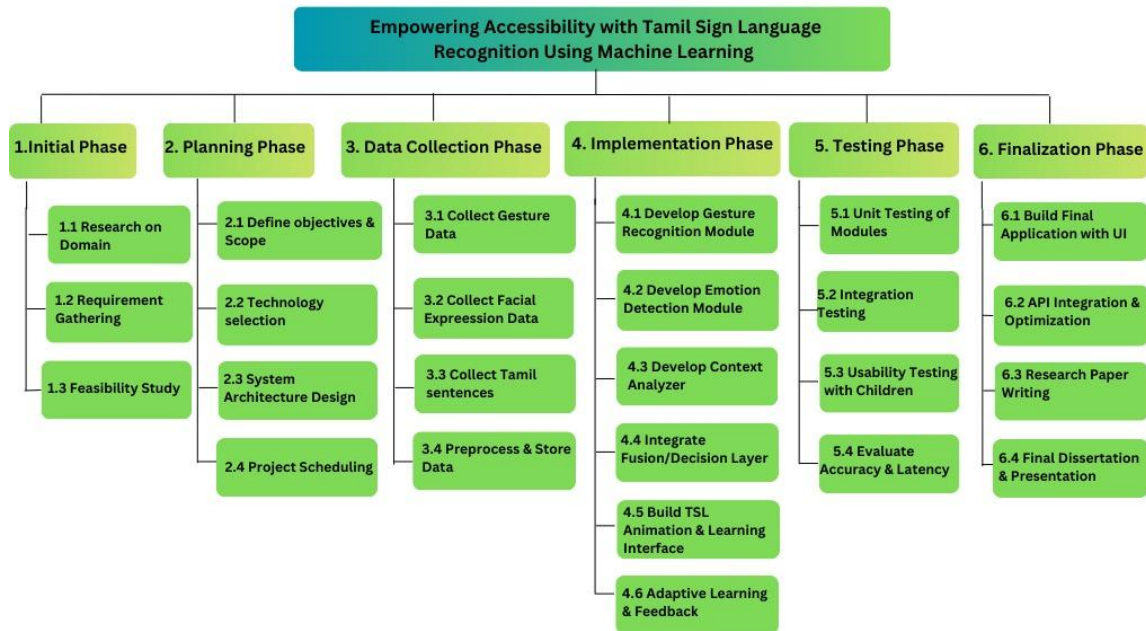


Figure 3.3 Work Breakdown Structure

3.5 Gantt Chart

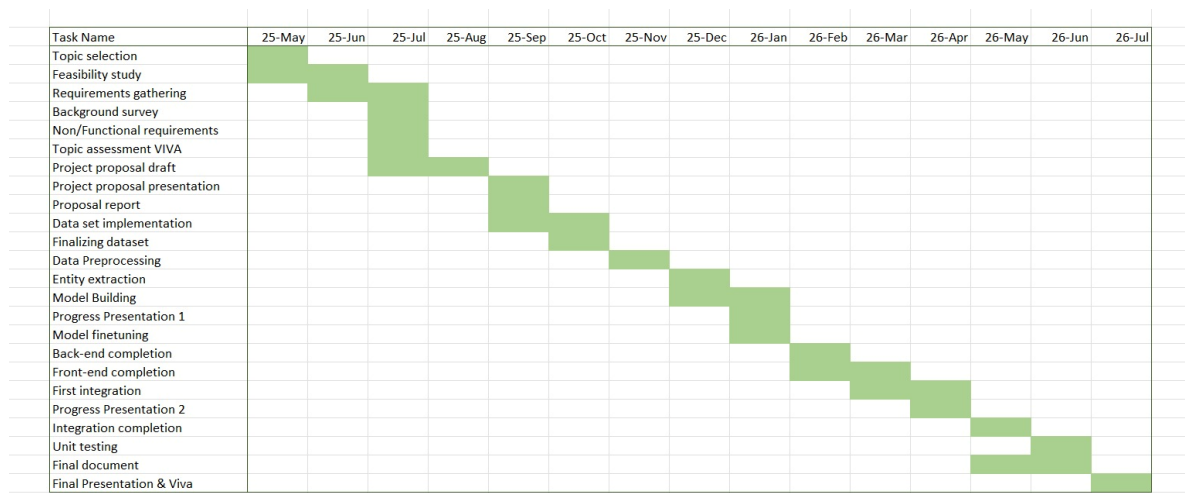


Figure 3.4 Gantt Chart

4 PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system should have the capability to analyze and recognize the facial expressions of the child in real time and recognize emotions such as happy, sad, angry, or neutral.
- The system should have the capability to recognize and classify the non-verbal noise, such as whimpers, murmurs, or crying, to recognize the emotional state of the child.
- The system should possess a fusion and mapping engine that combines facial and audio analysis results to produce a physical emotional output.
- The system should generate TSL avatar animation with text labels (e.g., "I am sad", "I am hungry") so that caregivers will know at once what the child wants.
- The system should provide real-time feedback by giving almost-instant responses such as "Correct" or "Try again" to support children and caregivers.
- The interface should be interactive and child-friendly, using simple navigation, vivid pictures, and avatars to enable easy and natural communication.
- The system must be capable of continuous use on normal equipment (laptop or tablet) and accommodate real-time application within homes, schools, and therapy facilities.

4.2 Non-Functional Requirements

- The facial emotion recognition and classification of non-verbal sounds should be high in accuracy so that they are able to offer consistent results.
- Fast and seamless error recovery should exist so that the system will be responsive and effective under real-time conditions.
- The interface should be colorful, intuitive, and child-like in nature so that it can be effortlessly operated by small children.
- The platform has to be device-agnostic, i.e., it has to be deployable on laptops, tablets, or the internet without any difference in reliability.

- The platform needs to ensure data security and privacy, protecting sensitive children information by storing it encrypted and anonymizing recordings.
- The platform needs to be robust and reliable, capable of running indefinitely in real-world settings such as classrooms, homes, or therapy clinics without needing to crash repeatedly.

4.3 System Requirements

- The system must be able to handle a stable internet connection for cloud storage, real-time processing, and deployment on cloud platforms like AWS or Google Cloud.
- The system must have sufficient processing power and storage to handle parallel facial emotion detection and non-verbal sound analysis seamlessly.
- The system must make use of relevant software tools and libraries, like Python, OpenCV, MediaPipe, Librosa, and Praat for backend processing and React.js for the frontend interface.
- The system should support the integration of APIs or custom modules for TSL avatar animations and text outputs that are expressive.
- The system should be able to maintain a secured database for processed features, logs of user interaction, and anonymized datasets of non-verbal sounds and facial expressions.

5 BUDGET AND BUDGET JUSTIFICATION

The total estimated budget for the Tamil Sign Language (TSL) Audio-to-Sign Conversion System is LKR 15,500, carefully distributed across four main categories. This allocation ensures the successful collection of data, development of the system, testing, and deployment in real-world contexts.

1. Data Collection - LKR 4,000

The system requires high-quality Tamil speech data and corresponding Tamil Sign Language gesture datasets for training and validation. Since TSL resources are limited, part of this budget will be used for curating datasets through collaboration with educators, deaf schools, and community experts. Additional costs may include transcription, annotation, and basic preprocessing of audio and gesture recordings.

2. Software Development - LKR 4,500

The development stage includes:

- Implementing the speech recognition module (local ML-based ASR trained on Tamil datasets).
- Training logistic regression models for mapping tokenized Tamil text to TSL gestures
- Developing a user-friendly API to connect the backend modules with the frontend interface.
- Designing the frontend application using React.js to visualize real-time gesture outputs.
- Backend integration through frameworks like Spring Boot with MySQL for database management

This budget ensures the availability of development tools, libraries, and potential costs for external APIs during prototyping.

3. Testing and Validation - LKR 4,500

Testing is crucial to evaluate system accuracy, performance, and usability. This includes:

- Accuracy checks on speech-to-text recognition under different accents and noisy conditions.
- Validation of gesture prediction outputs for linguistic correctness.
- User acceptance testing with members of the Tamil-speaking hearing-impaired community.
- Iterative improvements based on feedback from educators and accessibility experts.

This allocation also covers logistical expenses for user trials and validation sessions.

4. System Deployment – LKR 2,500

Deployment involves hosting the system on a secure and scalable platform. Hosting may include:

- Cloud deployment with API integration for easy accessibility.
- Providing access to schools, NGOs, and public organizations supporting hearing impaired communities.
- Basic maintenance and monitoring of the deployed system to ensure reliability.

The allocation ensures that the system is accessible to its target audience in real-world settings.

This budget allocation of LKR 15,500 strikes a balance between research rigor and resource efficiency. By prioritizing lightweight models and scalable software design, the system remains affordable while maintaining academic and practical value. The distribution emphasizes data quality, robust software development, extensive testing, and accessible deployment, ensuring that the Tamil Audio-to-TSL system is not only technically sound but also socially impactful.

Table 5.1 Estimated budget table.

Description	Total Amount (Rs.)
Data Collection	4000.00
Software Development	4500.00
Testing and Validation	4500.00
System Deployment	2500.00
Total	15500.00

6 APPENDICES

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Appendix -A Turnitin Report 1

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